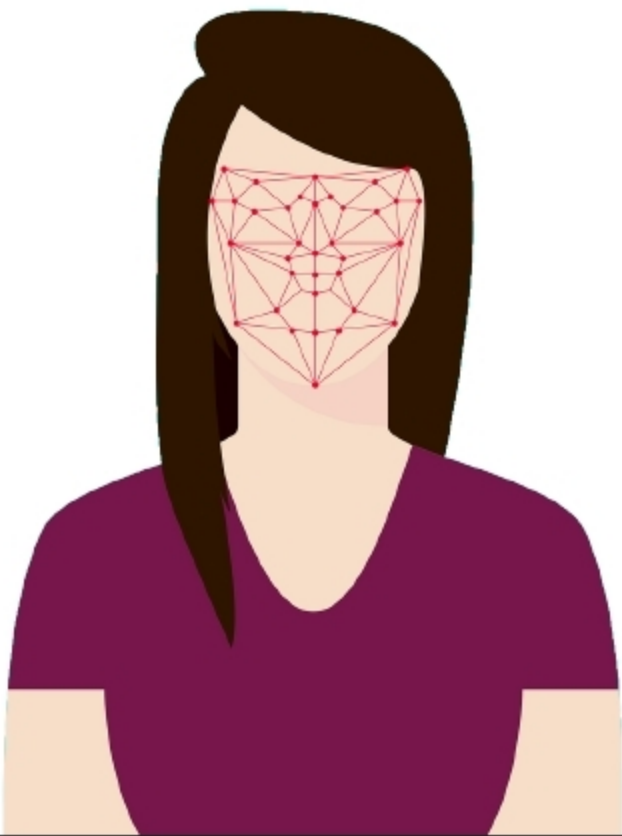


Using Facial Recognition for a Robust, Real-time Attendance System



There is a growing need to use technology to mark the attendance of students in the classroom – a process that is both automatic as well as efficient. In recent years, the problem of attendance management has been addressed through the use of standard biometrics like fingerprint or iris scanners. However, most proposed technologies are not too reliable. One solution is the intelligent application of facial recognition techniques iteratively. This article discusses an automated attendance management system that uses facial recognition.

Facial recognition comes under the category of biometric technologies like fingerprint and iris scanning. Facial recognition is capable of identifying a person from an image or from a video source. It is one of the best and simplest biometric technologies as it uses the facial features of the person only for identification. The main advantage of facial recognition compared to other biometric techniques is that it requires no contact as it captures the images of the person from a distance. This technology is being used by a lot of mobile companies, as well, as a secure yet password-less way to lock smartphones. Oneplus 5T and Apple

iPhone X are examples of smartphones that use facial recognition technology.

Face recognition can be used for many other purposes too, such as recording the attendance of students in colleges or schools, marking the presence of employees in companies, etc. In this article, we will mainly focus on using facial recognition for an attendance system.

The attendance system is widely used all over the world for various purposes and across diverse domains such as offices, schools, universities, etc. Marking the attendance of students in a classroom not only takes some effort but is also time consuming. Also, proxy attendance is always a possibility in classrooms where the number of

students is large, making it difficult for the faculty members to track the students who skip lectures.

Traditional methods of tracking the attendance of students are tedious. Conventionally, the teacher calls out the names of individual students who are supposed to be attending the class, and those present in the room respond; or the teacher passes around an attendance sheet for students to mark their attendance — both manual processes involve pen and paper and are not efficient.

Most educational institutions are concerned about their students' attendance at courses. Proper participation of students in the classroom leads to effective learning